

Julian Louis Ruocco

(484) 800-2952 | julianruocco@gmail.com | <https://julianruocco.com/>

Executive Summary

Aerospace Engineering senior graduating Spring 2027 and planning to begin M.S. study in Fall 2027. Seeking a Summer 2027 graduate-level systems engineering internship working on space systems, mission planning, or control systems.

Education

The Pennsylvania State University

B.S. Aerospace Engineering

Engineering Leadership Development Minor | Space Systems Engineering Certificate

May 2027

GPA 3.66/4.0

In Major GPA 3.82/4.0

Skills

Software & Tools: Python, MATLAB, Git, NumPy, Java, HTML, LabVIEW, KiCad, F Prime, Arduino

Simulation & CAD: STK, COMSOL, SolidWorks, CAD, digital twin modeling, MBSE, SysML

GNC & Hardware: GNC, ADCS, control systems, sensor and actuator trades, PID control, payload and hardware integration

Systems Methods: Systems engineering, requirements definition, trade studies, mission concepts, PDR, technical proposals

Work Experience

Mission Concept Intern, Space Dynamics Laboratory, Albuquerque NM

May - July 2026

- Developed a mission concept for a 6U CubeSat to identify and geolocate ocean plastics in the Pacific Ocean using onboard multispectral imaging and mission operations analysis.
- **Led ADCS trade studies that converted pointing and attitude-knowledge requirements into reaction-wheel performance targets and star-tracker sensitivity selections for the spacecraft.**
- **Used STK to optimize orbital designs for imaging-pass timing, ground-track coverage, and S-band antenna downlink performance within power budget and ground-contact constraints.**
- Coded a Python analysis pipeline that compressed Landsat multispectral imagery into spectral indices and scored candidate plastic signatures to assess detection feasibility before freezing requirements.
- Consulted with NASA & AFRL engineers to continually iterate the design to increase mission feasibility and relevance.

Research Projects

Control Systems Researcher, Astrodynamics Group of Penn State, State College PA

February-May 2026

- Modeled a 6 DOF Robotic Arm in Python using DH Parameters, NumPy, and simulated within Manim.
- Investigated Forward & Inverse Kinematics, Monte Carlo Method, and a Gaussian Mixture Model.

Honors PID Controller Project, AERSP 304 Dynamics and Control of Aerospace Systems

January-May 2026

- Independently developed a PID speed controller for a DC motor using an Arduino, actively capturing and analyzing real-time RPM telemetry within MATLAB.

Student Space Programs Laboratory, State College PA

Laboratory Lead & Chief Engineer

August 2025 - Present

- **Led 20 student engineers to win a total of \$125,000 in multiple phases of NASA's LunaRecycle Challenge.**
- Designed a novel modular recycling system (RECLAIM) for lunar missions using high power microwaves.
- Defined system requirements with all subsystems leads and integrated them into SysML linked to a Digital Twin which gathered live telemetry from pressure and temperature sensors to validate system requirements.
- Utilized Digital Twin to predict system behavior in off-nominal cases and extraterrestrial environments.
- Integrated and tested RECLAIM under vacuum conditions and temperatures exceeding 500 degrees Celsius.
- Fabricated a proof-of-concept microwave-assisted plastic pyrolysis system with less than \$500 and lab equipment.
- Directed laboratory initiatives, refined the training program, and taught 50+ student engineers systems engineering.

Research Lead, NASA's LunaRecycle Challenge (Phase 1)

December 2024 - May 2025

- Directed the research team for NASA's LunaRecycle Challenge (Phase 1), authoring key proposal sections.
- Created a MATLAB heat transfer model to gauge system temperature, improving energy efficiency by ~15%.

Member, Student Training Program

August 2023 - November 2024

- Applied systems engineering to design and build an atmospheric volatile gas detection payload for a suborbital rocket.
- Designed the electrical wiring of payload in KiCad and integrated the flight hardware onto a circuit board.

Research

Space Systems Researcher, Space Propulsion and Environments Laboratory, State College PA

May - August 2025

- Researched high power microwave systems applications for future lunar missions.
- Engineered a technical design and proposal for a lunar recycling system powered by a centralized microwave network.
- Integrated ISRU and catalysts into the design, enabling the recycling of plastics into rocket fuel precursors.